

Multiple Choice

1. **Answer:** A

Options: A) ROM; B) FPGA; C) PLA; D) PLD; E) EEPROM

Note: Correct option: A - ROM

2. **Answer:** B

Options: A) $f(t) = [e^{-2t} + 2.24e^{-2t}\sin(5t - 26.6^\circ)]u(t)$; B) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$; C) $f(t) = [e^{-2t} - 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$; D) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t + 26.6^\circ)]u(t)$

Note: Correct option : B - $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$

2. **Answer:** A

Options: A) 2; B) 8; C) 7; D) 12; E) 3

Note: Correct option: A - 2

3. **Answer:** A

Options: A) 4 meter²; B) 100 meter²; C) 50 meter²; D) 40 meter²; E) 16 meter²

Note: Correct option: A - 4 meter²

4. **Answer:** A

Options: A) either series or parallel.; B) in series.; C) in an alternating manner.; D) in series-parallel.; E) in anti series.

Note: Correct option: A - either series or parallel.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'AB = |76| |75| |168|'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '75%'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $V_{ab}(t) = -15te^{-4t} - 15e^{-4t} + 18e^{-2t}$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Number of isotopes.'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $1 - \cos t$ '.

Long Form Response

11. **Answer:** 601 lbm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 83.70%, 47.85 hp. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key